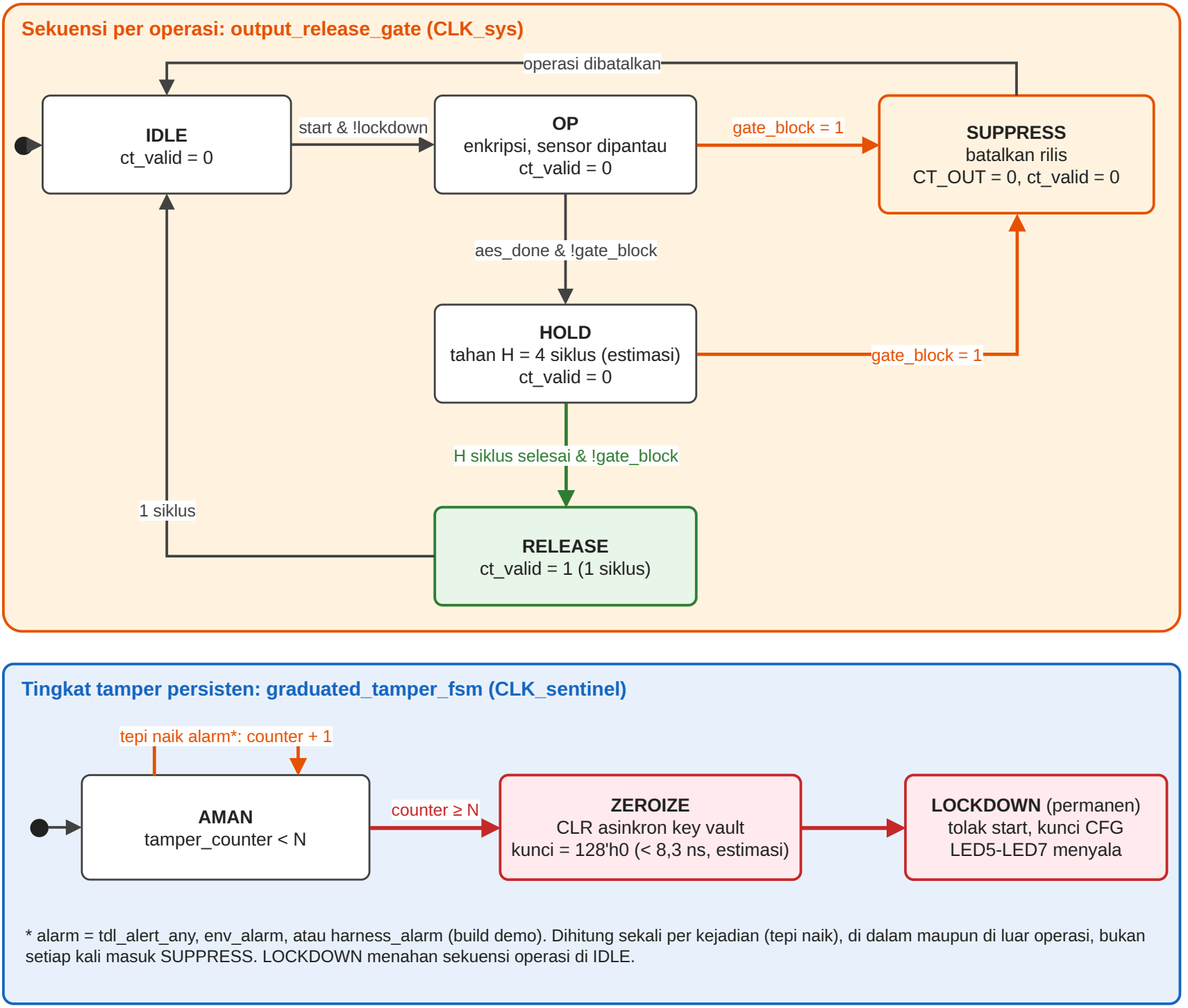
Sekuensi per operasi: output_release_gate (CLK_sys)
operasi dibatalkan
IDLE
ct_valid = 0
start & !lockdown
OP
enkripsi, sensor dipantau
ct_valid = 0
gate_block = 1
SUPPRESS
batalkan rilis
CT_OUT = 0, ct_valid = 0
aes_done & !gate_block
HOLD
tahan H = 4 siklus (estimasi)
ct_valid = 0
gate_block = 1
H siklus selesai & !gate_block
1 siklus
RELEASE
ct_valid = 1 (1 siklus)
Tingkat tamper persisten: graduated_tamper_fsm (CLK_sentinel)
tepi naik alarm*: counter + 1
AMAN
tamper_counter < N
counter ≥ N
ZEROIZE
CLR asinkron key vault
kunci = 128'h0 (< 8,3 ns, estimasi)
LOCKDOWN (permanen)
tolak start, kunci CFG
LED5-LED7 menyala
* alarm = tdl_alert_any, env_alarm, atau harness_alarm (build demo). Dihitung sekali per kejadian (tepi naik), di dalam maupun di luar operasi, bukan setiap kali masuk SUPPRESS. LOCKDOWN menahan sekuensi operasi di IDLE.